

11/19/2022 Pipe Flow Pre-Lab

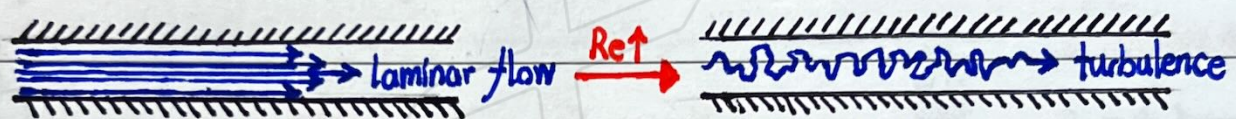
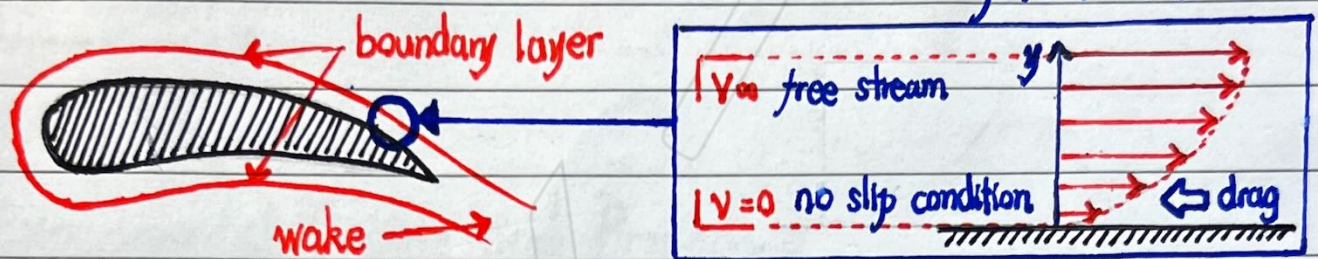
Introduction

■ effect of solid walls on fluid flows

④ the interaction of flows with curved surfaces

④ effects of the boundary layer generated

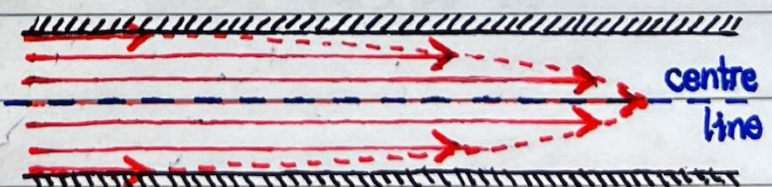
better understand
real world
phenomena



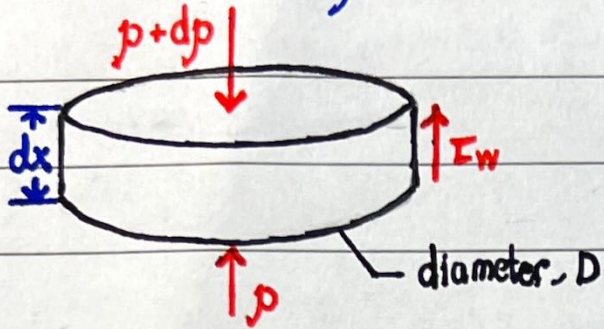
- examine the transition from laminar to turbulent
 - measuring the pressure drop along the length of the pipe
 - allow us to compute the shear stress, τ
- investigate the C_f skin friction coefficient varies with Re
- deduce the Re_{crit} critical Reynolds number
- compare results to theory and past findings

Theory

fully developed pipe flow



11/20/2022 Pipe Flow Pre-Lab



$$\frac{\pi D^2}{4} \frac{dp}{dx} dx = \pi D \tau_w dx$$

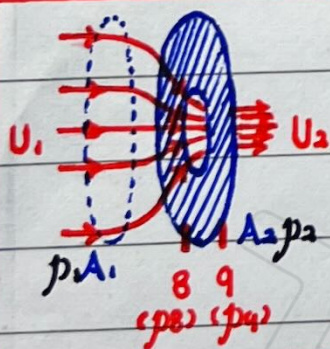
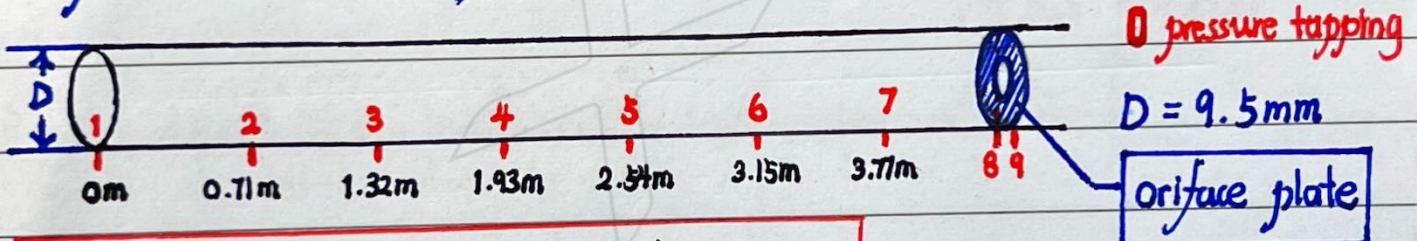
$$\downarrow$$

$$\tau_w = \frac{D}{4} \frac{dp}{dx}$$

$$Re_D = \frac{\rho V D}{\mu}$$
$$C_f = \frac{\tau_w}{\frac{1}{2} \rho V^2} = f(Re_D)$$

$$C_f = \frac{P}{\frac{1}{2} \rho v^2} \frac{dp}{dx} = \frac{\Delta p}{\frac{1}{2} \rho v^2} \frac{D}{4L} = f(Re)$$

Experimental Facility and Procedure



$$A_2 < A_1 \xrightarrow{Q=\rho u A} U_2 > U_1$$

$$p_1 + \frac{1}{2}\rho U_1^2 = p_2 + \frac{1}{2}\rho U_2^2$$

$$\hookrightarrow p_1 < p_2$$

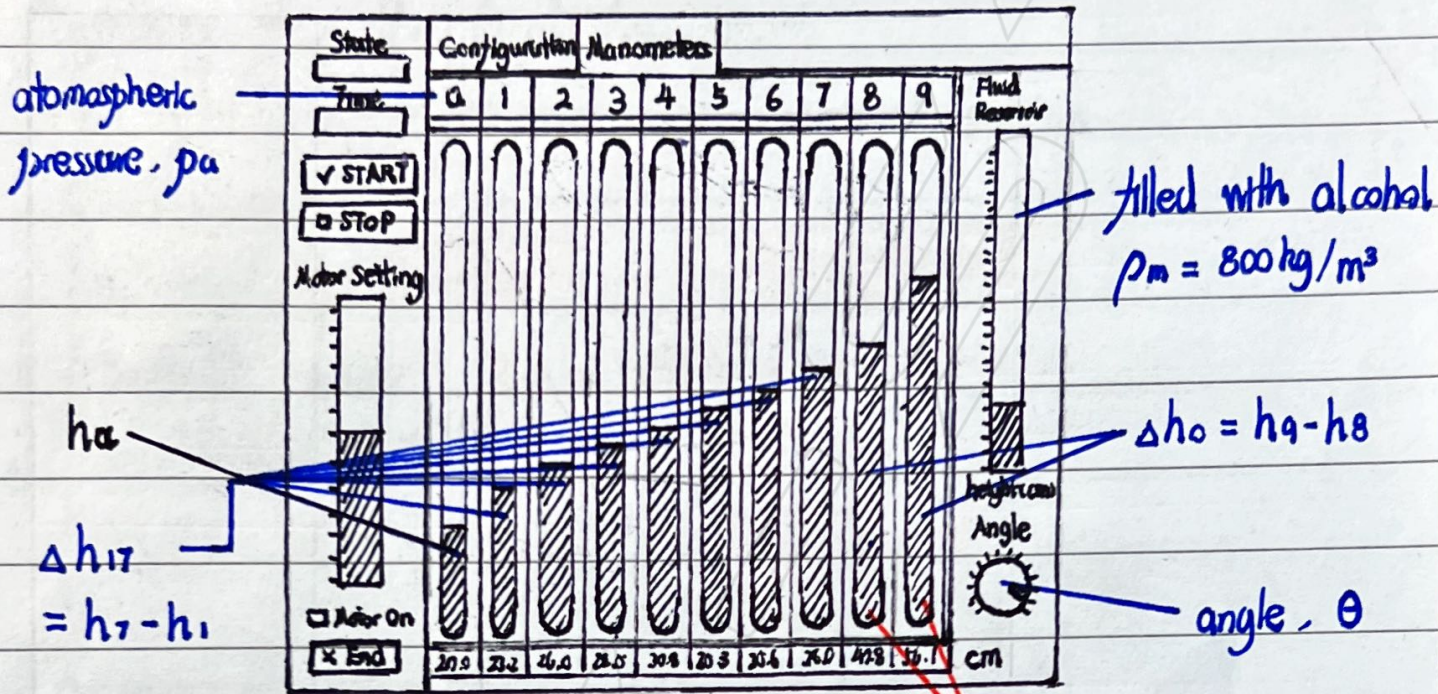
$$\hookrightarrow p_9 > p_8 \text{ 突破梗阻}$$

$$\rightarrow k_{lam} = 0.10$$
$$\rightsquigarrow k_{turb} = 0.08$$

$$\rightarrow \frac{1}{2} \rho_{\text{air}} V^2 = k \Delta p_0 = k (p_9 - p_8)$$

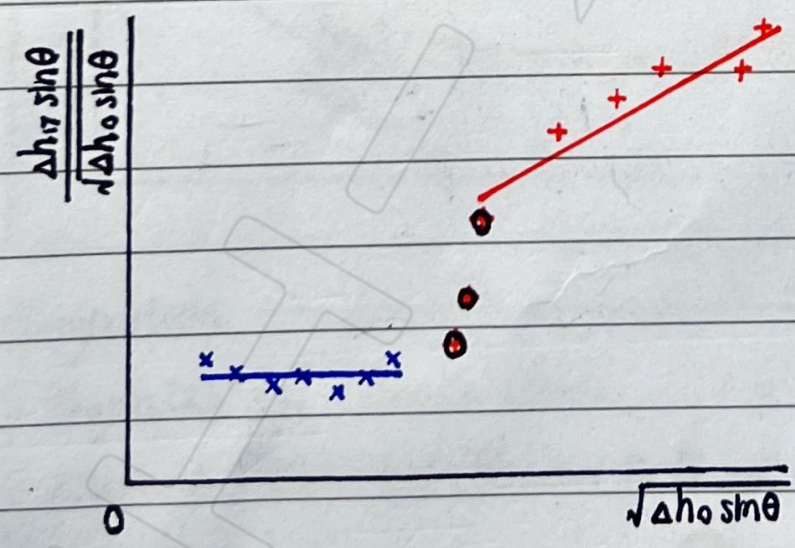
determine the mean velocity in the flow

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$$\Delta p_o = \rho g \Delta h = \rho_m g \Delta h_o \sin \theta$$

$$\frac{1}{2} \rho_a V^2 = k \Delta p_o$$



- Laminar Test Cases
- Transition Test Cases
- + Turbulent Test Cases

Running Plot

- isolate transition
- determine the distribution of points

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Preparing for the Oral Assessment

24h after lab $\rightarrow$ 30 min oral assessment session

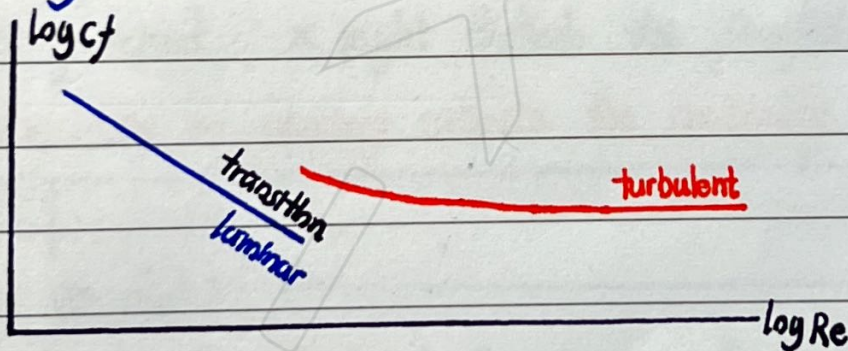
Calculation

$$\text{velocity } V = \sqrt{\frac{2k\Delta p_0}{\rho_a}} = \sqrt{\frac{2k\rho_m g \Delta h_0 \sin \theta}{\rho_a}}$$

$$\text{Reynolds number } Re_D = \frac{\rho V D}{\mu} = \frac{\rho_a V D}{\mu_{\text{air}}}$$

$$\begin{aligned} \text{skin-friction coefficient } C_f &= \frac{\Delta p}{\frac{1}{2}\rho_a V^2} \frac{D}{4L} \\ &= \frac{\rho_m g \Delta h_0 \sin \theta}{k \rho_m g \Delta h_0 \sin \theta} \frac{D}{4L} \\ &= \frac{\Delta h}{k \Delta h_0} \frac{D}{4L} = \frac{1}{4k} \left[\frac{d(\frac{h}{h_0})}{d(\frac{x}{L})} \right] \text{ gradient} \end{aligned}$$

Plotting



Determine the slope of the straight line

Comparison

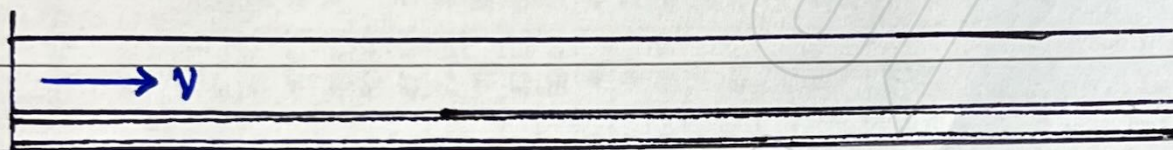
Hagen [2] and Poiseuille (1840) $C_f = \frac{16}{Re_D}$

Nikuradse (1932) and Stanton & Pannell (1914)

$$\frac{1}{\sqrt{C_f}} = -0.4 + 4 \log (Re_D \sqrt{C_f})$$

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- $P_{\text{atmosphere}} = 984.9 \text{ mbar} = 98990 \text{ Pa}$
- $T = 24.2^\circ\text{C} = 297.35 \text{ K}$



* pressure tupper 4 do not exist in equipment
ignore the value record by tapping 4

* value is unstable at first few seconds in steady flow

* value cannot keep in a constant value in unsteady flow
just approximate it at average

Other values

$$\mu_{\text{air at } 24.2^\circ\text{C}} \approx 1.84 \times 10^{-5} \text{ kg/ms}^2$$

$$\rho_a = \frac{p_a M}{RT} = 1.15967492 \text{ kg/m}^3$$

$$\rho_m = 800 \text{ kg/m}^3$$

$$D = 9.5 \text{ mm} = 9.5 \times 10^{-3} \text{ m}$$

$$\text{fluid reservoir height} \\ h = 20 \text{ cm}$$

* μ_{air} is estimated value from diagram standard

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Number of attempts	Motor Setting (V)	Angle (degrees)	Fluid Reservoir height (cm)	Pressure Tapping															
				Position x (m)	a	1	2	3	4*	5	6	7	8	9					
1				h (cm)															
2																			
3																			
4																			
5																			
6																			
7																			
8																			
9																			
10																			
11																			
12																			
13																			
14																			
15																			
16																			
17																			
18																			
19																			

Ignore pressure tapping cause it is not exist

* Ignore pressure tapping cause it is not exist

* Data need to be processed again with the error occurs at ground state

11/21/2022 Pipe-flow During Lab

Number of attempts	Motor Setting (V)	Angle (degrees)	Fluid Reservoir height (cm)	Pressure Tapping									
				x (m)	error (m)	h (m)							
1													
2													
3													
4													
5													
6													
7													
8													
9													
10													
11													
12													
13													
14													
15													
16													
17													
18													
19													

* Some value might be invalid due to the inaccurate estimate from reading

11 / 21 / 2022

Pipe-flow Post-Lab

[illegible]

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Find gradient
to estimate
the C_f

$C_f =$
 $(\frac{dh}{dx})_{\text{gradient}}$

Laminar and
turbulent flow
have much
higher gradient
than transition
flow, and
laminar flow
have very
large-disturbed
data point

ESTIMATE THE GRADIENT (1000*ACTUAL GRADIENT)

$h_n - h_1 / \Delta h_0$

$x_n - x_1 / 1000D$

11/21/2022 Pipe-flow Post-Lab

Number of attempts	Motor Setting (V)	Type of Flow	V (m/s) $(2k\Delta p_0/\rho_s)^{1/2}$	Re _D $\rho_s V D / \mu$	T _w (Pa) $(D/4)(dp/dx)$	C _f $(1/4k)\text{gradient}$	C _f comparison		
							Hagon and Poiseuille 16/Re _D	Nikuradse and Stanton & Pannell $1/(Cf)^{1/2} = -0.4 + 4\log(Re_D(Cf)^{1/2})$	Δ
1	0								
2	9.3								
3	9								
4	8								
5	7								
6	6								
7	5								
8	4								
9	3								
10	2								
11	1								
12	5								
13	5.46								
14	0.5								
15	1.25								
16	1.5								
17	0.35								
18	0.62								
19	0.72								

* Attempts 14, 18, 19 is invalid caused by the measurement error

Type of flow can be determined through the running plot in next page →

$$k_{\text{lam}} = 0.10$$

$$k_{\text{turb}} = 0.08$$

Assume that

$$k_{\text{tran}} = 0.08$$

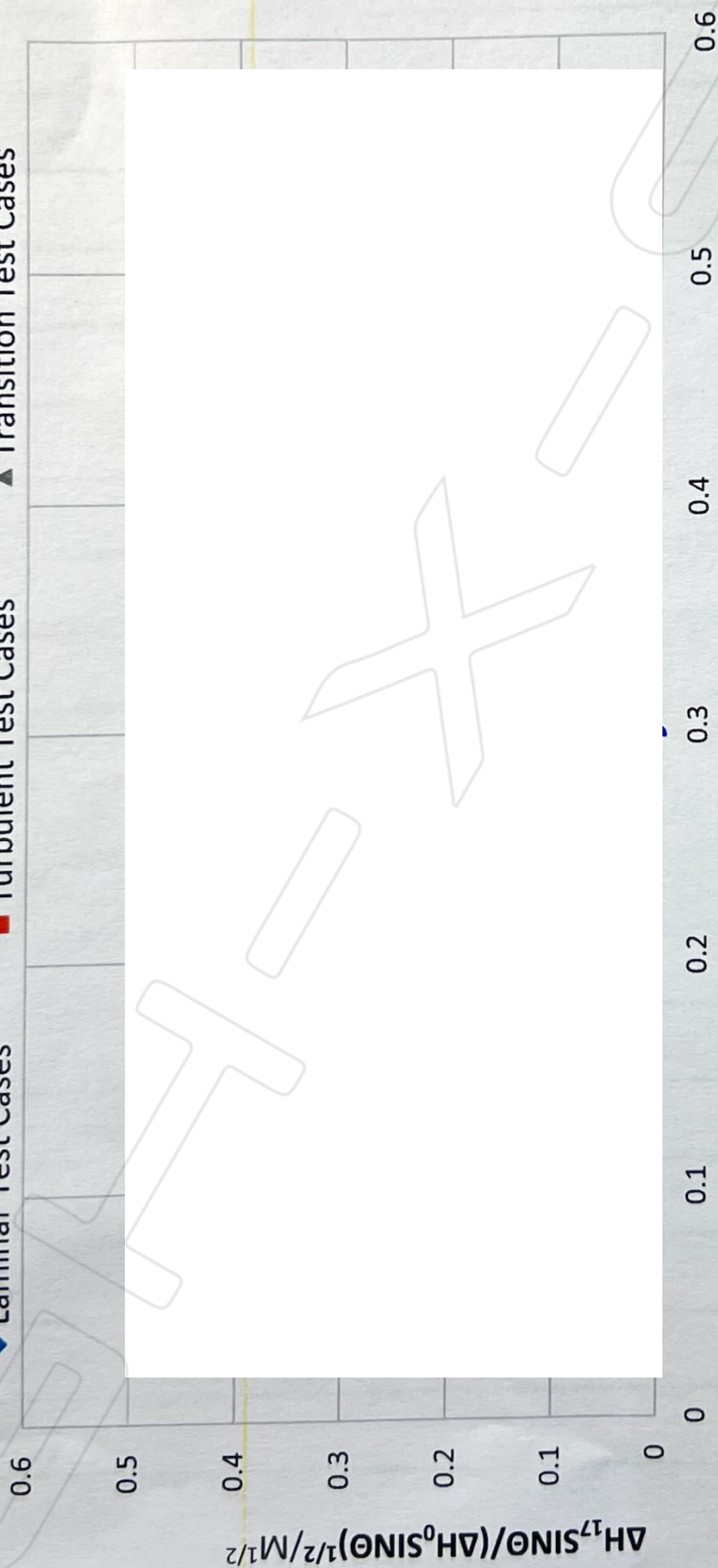
Δ might caused by the energy dissipation

Δ might caused by the measurement error

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RUNNING PLOT

◆ Laminar Test Cases ■ Turbulent Test Cases ▲ Transition Test Cases

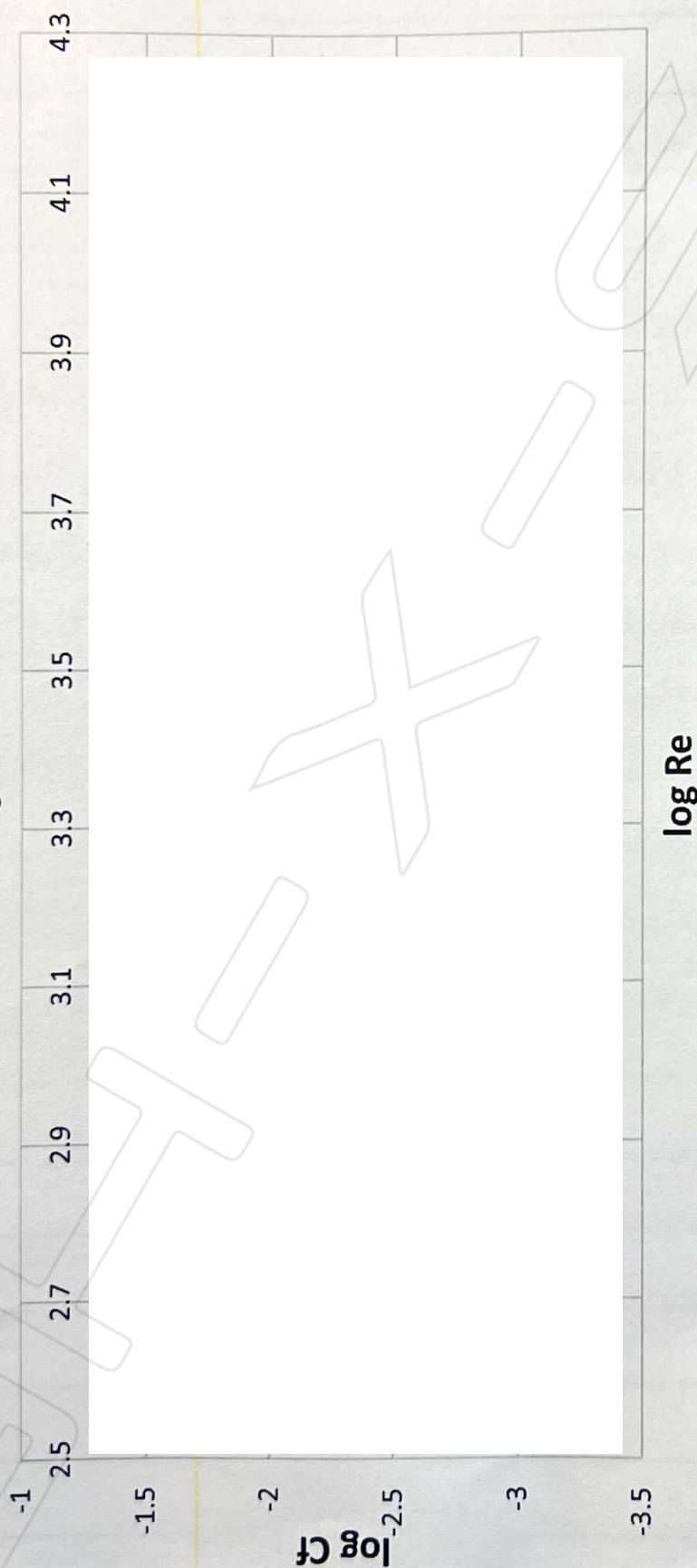


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VARIATION OF SKIN FRICTION COEFFICIENT AS A FUNCTION OF

REYNOLDS NUMBER

◆ Laminar ■ Turbulent ▲ Transition × Hagon and Poiseuille * Nikuradse and Stanton & Pannell



× Hagon and Poiseuille
use to calculate the
laminar flow
slightly higher than
experimental value

* Nikuradse and
Stanton & Pannell
use to calculate the
turbulent flow
slightly higher than
experimental value

The gap is shown
by the Δ in the
graph in previous
page